

# RAG for Legal Documents Information Retrieval

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Reasoning AI Chatbot

# Project Layout

## Phase1: Setup UI with Streamlit

- Setup Upload PDF functionality
- Chatbot Skeleton (Question & Answer)

## Phase2: Setup Vector Database (Memory for LLM)

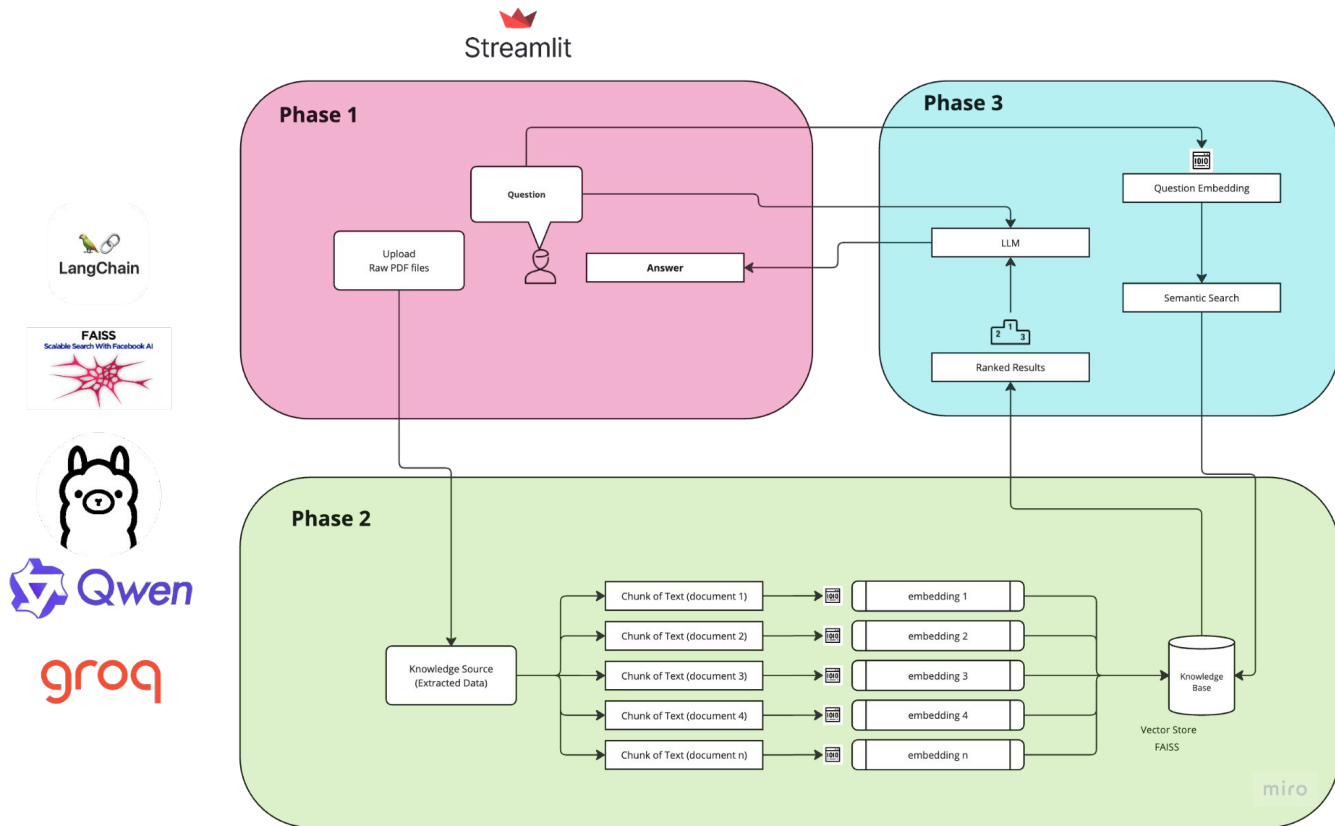
- Upload & Load raw PDF(s)
- Create Chunks
- Setup Embeddings Model (Use Ollama or any other OS model supported by groq)
- Index Documents \*\*Store embeddings in FAISS (vector store)

## Phase3: RAG Pipeline (Connect Memory with LLM)

- Setup LLM (Use Ollama/QWEN with Groq)
- Retrieve Docs
- Answer Question

## Phase4: Connect Everything Together

# Technical Architecture



# Tools and Technologies

- Ollama/Qwen/Mistral (Best Free CoT Reasoning Model)
  - With GROQ
- Ollama (Local LLM hosting/managing platform)
- Langchain (AI Framework for LLM applications)
- RAG, What is RAG and why do we need it?
- Streamlit (For Chatbot UI)
- FAISS (Vector Store)
- Pdfplumber and langchain loaders

# RAG–Retrieval Augmented Generation

## Basics:

- Models are trained up to a limit
- No proprietary information access
- Answers from specific sources
- Reduce Hallucinations
- Vector Embeddings

## Vector Embeddings:

- Numerical representations of words or phrases
- Capture meanings and relationships
- Helps machine learning models understand text effectively

